

SYNAPSE

A Computational Disease Dynamics Company

DOCUMENT	North Star — 3-Year Strategic Roadmap
VERSION	3.0 (Independent Path)
AUTHOR	Liban Britt
STATUS	Active — Source of Truth
KEY SHIFT	UW as accelerant not dependency; OLIVES-first; deeper plan
AUDIENCE	Founder, Co-founders, AI Collaborators

"We model disease as a hidden dynamical system observed through noisy signals and modified by intervention. We begin with diabetic retinopathy, prove the architecture on open data we can access today, and build toward a general computational disease dynamics platform. We depend on no single institution — we earn every advantage through evidence."

What Changed in Version 3.0

Three structural shifts from v2, plus gap coverage:

UW is now an accelerant, not a dependency. The entire plan is designed to work WITHOUT institutional affiliation. OLIVES is open-download (CC BY 4.0), so the dynamics proof-of-concept starts now, solo. UW (if pursued) compresses timelines and eases controlled-data access — but nothing critical-path depends on it. This de-risks the plan enormously.

OLIVES-first execution. OLIVES (96 eyes, fundus+OCT, 16 biomarkers, treatment timing, ~2yr follow-up) is the first real Synapse-shaped dataset. The dynamics PoC begins with a data feasibility audit, then sequence modeling — all on data downloadable today.

Deeper high-level plan. Explicit 'what we want to achieve' for each phase, the full capability ladder, and the logic connecting each phase to the next.

Gaps covered. IP strategy, runway math, deeper competitor analysis, the data feasibility audit as literal first task, and multiple independent data-access paths.

Part I — What We Are Building and Why

1. The Company in One Page

Synapse is a computational disease dynamics company. The core product, at maturity, is an intervention-aware latent state-space model of disease progression — a 'physics engine' for how disease evolves under treatment. We start with diabetic retinopathy because it has the best data infrastructure in medicine, then expand to other imaging-rich progression diseases (AMD, glaucoma) and eventually toward a general platform.

What we sell (eventually): trajectory forecasting, counterfactual treatment simulation, virtual trial design, and validated synthetic cohorts — to pharma, biotech, CROs, and researchers. **What we are NOT:** a DR screening tool, a synthetic-data vendor, a clinical chatbot, or a diagnostic device. Those are crowded or commoditized. We model where disease is going, not what it is right now.

2. The Core Insight (Why This Is Different)

Medicine stores observations — scans, notes, labs, treatments — but does not explicitly model the underlying disease process. Good doctors implicitly perform Bayesian inference: observe noisy signals, estimate hidden disease state, intervene, update beliefs, repeat. We formalize that computationally.

Observations are not the disease. The disease is hidden. A retinal image, a lab value, a symptom — these are noisy projections of an underlying biological process. Our entire framework rests on inferring that hidden state and modeling how it evolves over time and under intervention. This is the load-bearing idea. Remove it and we collapse into 'another healthcare AI startup.' Keep it and we have a coherent probabilistic framework that subsumes forecasting, simulation, and synthetic generation.

3. What We Want To Achieve (The Capability Ladder)

Each rung is a concrete capability. Each enables the next. We do not skip rungs. This ladder IS the strategy — everything else is timing and resourcing.

Rung	Capability	Proves	Phase
1	Strong retinal representation	We can encode disease signal	Y1 (public)
2	Temporal modeling on real sequences	Hidden state evolves learnably	Y1 (OLIVES)
3	Scaled progression forecasting	It works beyond a toy cohort	Y2 (controlled)
4	Intervention-aware dynamics	We model treatment effects	Y2-3
5	Counterfactual simulation	What-if reasoning works	Y3
6	Validated synthetic cohorts	Generation preserves causality	Y3+
7	Multi-disease platform	The engine generalizes	Y3-4

8	Disease dynamics infrastructure	Category leadership	Y4-7
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The discipline: We are on Rung 1-2 today. We do not claim Rung 3+ until the data and evidence support it. The vision is all eight rungs; the execution is the next rung.

4. Honest Probability Assessment

We do not pretend this is easy or certain. The realistic distribution, assuming disciplined execution and the independent (non-UW-reliant) path:

Outcome	Probability	Timeline
Working prototype (public + OLIVES)	80%	Month 6
Publishable temporal-modeling result	50%	Month 9-12
Controlled-data access secured	55%	Month 9-18
Dynamics thesis validated at scale	25%	Month 18-24
\$100M+ company	5-10%	Year 4-5
Unicorn (\$1B+)	2-5%	Year 5-7

Probability of the prototype rose to 80% (from 75%) because OLIVES removes the data-access dependency for Year 1. The biggest remaining gate is Rung 3 — scaling beyond OLIVES — which requires controlled data but no longer blocks the Year 1 proof. **A 2-5% shot at a billion-dollar outcome is an extraordinary bet. Our near-term mission is to earn the right to keep pursuing it.**

5. The Scientific Hypothesis (Falsifiable)

H1 (Architecture): Disease progression can be modeled as a latent state-space system where hidden state evolves via learnable transitions and observations are noisy projections. Prototyped on OLIVES in Y1; validated at scale on controlled data in Y2.

H2 (Intervention): With treatment history, the system recovers intervention-aware dynamics enabling counterfactual reasoning. OLIVES has treatment timing — partial test possible in Y1; full validation Y2-3.

H3 (Generation): A validated dynamics model generates synthetic cohorts preserving causal structure. Y3+.

What Year 1 Can / Cannot Prove (Honest Boundaries)

Claim	Year 1 Status
Strong retinal representation learning	YES (public data)
Architecture functions on real temporal data	YES (OLIVES — this is the win)
Treatment timing can condition the model	PARTIAL (OLIVES has it; small n)
Clinically plausible trajectories	PARTIAL (with caveats)

Progression modeling validated at scale	NO (needs controlled data, Y2)
Causal treatment effects proven	NO (needs more treatment data, Y2)
Synthetic cohort validity	NO (needs validated dynamics, Y3)

The honest Year 1 claim: 'We built an open-data temporal prototype demonstrating that multimodal retinal disease states can be modeled over time, including treatment information.' Not 'we solved DR progression.' The honest claim is still publishable and still anchors partnership conversations.

Part II — The Independent Path & Data Strategy

The defining principle of v3: **no single institution is on the critical path**. We build with data we can access today, and we pursue MULTIPLE independent routes to the controlled data needed for scale. UW, if pursued, is one accelerant among several — not a dependency.

6. Why We Are Not UW-Reliant

An earlier draft leaned on UW affiliation (Jan 2027) to solve data access. That created a single point of failure and tied the company's timeline to a graduate program. v3 removes that dependency:

- OLIVES is open-download (CC BY 4.0).** Year 1 dynamics PoC needs no affiliation, no IRB, no partnership. Start today.
- UK Biobank accepts commercial/independent researchers.** Public-interest health research eligibility is not limited to universities. ~£9K Tier 3, coverable by grants.
- EyePACS solicits partnerships directly.** A narrow commercial pilot does not require academic cover.
- Grants (SBIR/NEI/ARPA-H) fund independent founders.** Small Business programs are designed for companies, not universities.
- UW becomes optional upside.** If pursued, it eases UW Medicine collaboration, IRB, and co-authorship. But the plan succeeds without it.
- UW posture:** Treat any UW affiliation as a bonus accelerant. Build relationships (e.g., Aaron & Cecilia Lee's retinal AI lab) opportunistically. But never let a sprint, a milestone, or a funding decision hinge on it.

7. Dataset Reality

Open — usable now, solo:

Dataset	Size	Longitudinal	Role
OLIVES	96 eyes; 1,268 fundus + ~49 OCT each; 18,155 markers; treated as 1 PoC	Yes	Dynamics PoC — the first Synapse-shaped data
EyePACS public	~90K images	No	Encoder pretraining, grading baselines
Messidor / -2	~1.2K / 1.7K	No	External classification validation
FGADR	2,842 images	No	Lesion-aware auxiliary supervision

Controlled / partner — multiple independent routes for scale:

Dataset	Scale	Independent Access Route
EyePACS private	575K eyes; 28.9K w/ 2yr outcomes	Commercial partnership (contact@eyepacs.org)
UK Biobank	100K OCT participants	Direct application — commercial eligible, ~£9K

ACCORD EYE	2,856 patients, 4yr	BioLINCC request (confirm raw images)
DRCR / Jaeb	Protocolized trials	Partnership outreach
Single retina clinic	Varies	Direct pilot — narrow de-identified extract

8. The OLIVES Feasibility Audit (Literal First Task)

Before any modeling, we answer whether OLIVES has the temporal structure to support dynamics learning. This protects against the most common failure: building a beautiful latent ODE on data too thin to support the claim. Four questions, in order:

Q1 — File structure. How are eyes, visits, fundus, and OCT organized on disk?

Q2 — Temporal depth. Per eye, how many DISTINCT timestamped visits? Histogram: how many eyes have 2, 3, 5, 10+ visits?

Q3 — Alignment. Can imaging timepoints align with the 16 biomarkers AND the injection/treatment timeline on a shared clock?

Q4 — Usable subset. Largest clean longitudinal subset — eyes with enough timestamped visits for sequence modeling?

Decision criterion: If most eyes have ≥ 4 timestamped visits, a latent ODE is viable. If temporal depth is thin (2-3 visits), scope the claim to short-horizon prototyping and rely on simpler temporal baselines. Either is publishable — but we decide BEFORE modeling, not after.

9. Two-Track Acquisition Plan

Track 1 — Build now (solo, Months 1-6):

- OLIVES feasibility audit + per-eye sequence dataset
- RETFound encoding of OLIVES + EyePACS
- Encoder validation on Messidor/-2; FGADR auxiliary tasks
- Latent dynamics prototype on OLIVES sequences
- Treatment-conditioning experiments (OLIVES has injection timing)
- Preprint: open-data temporal prototype

Track 2 — Secure scale data (parallel, Months 1-18):

- Month 1: UK Biobank application started (independent eligibility)
- Month 2-3: EyePACS partnership inquiry (narrow ask + value prop)
- Month 3: BioLINCC ACCORD request (confirm raw imaging)
- Month 4-6: Single retina-clinic pilot outreach (local Seattle-area clinics)
- Month 6: ≥ 1 controlled path confirmed or denied
- Optional: UW Medicine collaboration if/when affiliated (bonus, not required)

Part III — Architecture (Full System)

The complete system has seven tiers. The MVP is Tiers 1-4 (minimal), built on OLIVES + public data. Tiers 5-7 emerge in Years 2-3. JEPA is a technique within Tier 2, not the brand. HAWVA scales from a simple uncertainty layer (MVP) to a full Bayesian fusion brain (Year 3).

Tier 1 — Data Ingestion

Ingest asynchronous, sparse, irregular observations; normalize; align timelines. MVP: OLIVES + EyePACS loaders.

MVP

Tier 2 — Encoder Layer

RETFound backbone (pretrained); JEPA-style predictive fine-tuning on top (Y1+); clinical + temporal encoders. JEPA is optional enhancement, not load-bearing.

MVP (RETFound); JEPA Y1+

Tier 3 — HAWVA Belief State

Persistent probabilistic belief over hidden disease state. MVP: uncertainty + calibration. Y2: multimodal fusion. Y3: full Bayesian brain.

Minimal in MVP

Tier 4 — Disease Dynamics Engine

The scientific core: $\text{state}(t+dt) = f(\text{state}(t), \text{treatment}(t), \text{patient}, \text{noise})$. Latent ODE on OLIVES (MVP); treatment-conditioned + physician-policy (Y2).

Latent ODE in MVP

Tier 5 — Active Inference

Recommend optimal next observation to reduce uncertainty. Decision support without treatment recommendation (stays out of SaMD early).

Y2-3

Tier 6 — Simulation & Generation

Trajectory forecasting, counterfactual simulation, synthetic cohorts, virtual trials.

Y2-3

Tier 7 — Products & APIs

REST/GraphQL API, SDK, dashboards, pharma integration, audit logging.

Core Equation

*The entire company reduces to whether this is learnable: **$\text{state}(t + dt) = f(\text{state}(t), \text{treatment}(t), \text{patient_factors}, \text{noise})$** . Everything — encoders, fusion, simulation, synthetic generation — is infrastructure around learning f .*

Part IV — Execution Roadmap

10. Year 1 — Prove the Architecture (Independent)

What we want to achieve: Climb Rungs 1-2 of the capability ladder. Demonstrate strong retinal representation AND that hidden disease state evolves learnably on real temporal data (OLIVES). Secure at least one path to scale data. Produce a credible preprint. All achievable solo, without affiliation.

Year 1 North Star Metric

By Month 12: a preprint demonstrating (a) strong retinal representation learning and (b) a functioning temporal dynamics prototype on OLIVES including treatment information, PLUS at least one controlled-data pathway approved or in late-stage review.

Q1 (Mo 1-3): Foundation + Feasibility

- OLIVES download + feasibility audit (the 4 questions)
- Reproducible infra: Git, W&B, clean repo, env specs
- EyePACS + Messidor pipeline; RETFound integration
- Baselines: logistic, GRU-D, T-LSTM, Cox survival
- Track 2: START UK Biobank application

Q2 (Mo 4-6): Dynamics Prototype

- Per-eye sequence dataset from OLIVES
- Latent ODE prototype on OLIVES sequences
- Treatment-conditioning experiments (injection timing)
- Encoder external validation (Messidor-2)
- Uncertainty + calibration
- Track 2: EyePACS partnership email; ACCORD request

Q3 (Mo 7-9): Enhance + Write

- JEPA-style fine-tuning on RETFound (compare vs frozen)
- Clinical encoder (OLIVES biomarkers)
- Draft preprint
- Track 2: retina-clinic pilot outreach

Q4 (Mo 10-12): Publish + Position

- Submit preprint to arXiv
- Submit to venue (MICCAI, NeurIPS, JAMA Ophth)

- First investor/partner conversations
- Gate 3 decision; secure scale-data path

11. Year 2 — Validate at Scale

What we want to achieve: Climb Rungs 3-4. Validate the dynamics thesis on real controlled longitudinal data. Add multimodal fusion and intervention-aware modeling. Make first business steps — seed funding or a substantial pilot.

North Star: By Month 24 — validated latent dynamics on controlled longitudinal data, multimodal HAWVA working, treatment-aware modeling demonstrated, seed closed or significant pilot in motion.

Q5 (Mo 13-15)

- Integrate controlled data (UK Biobank / EyePACS / clinic pilot)
- Replicate Y1 results at scale
- Multimodal HAWVA (image + clinical + temporal)
- Recruit technical co-founder or senior advisor

Q6 (Mo 16-18)

- Treatment-conditioned transition model
- Validate intervention effects vs clinical literature
- First API endpoints
- Close seed (\$500K-\$2M) or major grant

Q7 (Mo 19-21)

- First paid pilot or grant partnership
- Begin AMD model (second disease)
- Hire 1-2 if funded
- Second-stage paper

Q8 (Mo 22-24)

- AMD model in pilot
- Robust API + SDK
- Active inference R&D;
- Series A prep

12. Year 3 — Platform & Partnerships

What we want to achieve: Climb Rungs 5-7. Counterfactual simulation, validated synthetic cohorts, multi-disease platform. First real pharma revenue. Category-leader positioning.

North Star: By Month 36 — multi-disease platform (2-3 validated models), \$1-5M ARR or comparable funding, 5-10 person team, clear Series A momentum.

Platform

- DR + AMD validated; Glaucoma pilot
- Full HAWVA Bayesian fusion
- Active inference decision support
- Synthetic cohort generation
- Counterfactual simulation engine

Business

- 2-3 pharma partnerships active
- \$1-5M ARR
- Series A (\$10-25M) or strategic partnership
- Clinical advisory board

Science

- 3-5 peer-reviewed publications
- Open-source benchmarks
- Category-leader recognition
- Top-venue talks

13. AI-Accelerated Execution

Building with AI compresses some phases dramatically — but not all. Knowing the difference is itself a competitive edge.

AI accelerates: code generation (5-10x), debugging, documentation, literature review, experiment scaffolding, writing (3-5x).

AI does NOT accelerate: data acquisition (human relationships, legal agreements), clinical validation (clinicians' real time), scientific judgment (what to validate, what to claim), trust building, real-world testing (models work or don't), regulatory navigation, and the patience to sit with ambiguous results.

The three disciplines: (1) Use AI for breadth, human judgment for depth. (2) Verify everything — AI hallucinates; every claim, link, and reference is checked. (3) Maintain scientific rigor — AI will happily build a beautiful model that doesn't generalize; evaluation discipline must come from you.

Part V — Business, IP & Runway

14. Customers & Revenue

Segment	Use Case	Pricing	Timeline
Large Pharma	Trial design, RWE, drug dev	\$500K-\$5M/yr	Y2-4
Biotech (rare disease)	Synthetic cohorts, virtual trials	\$100K-\$1M/proj	Y3+
CROs	Bundle into trial services	Wholesale + rev share	Y3-4
Academic / research	Disease research, publications	Discounted/free	Y1-2
Regulators	RWE, surveillance	Complex/long	Y4-5+

Mature revenue mix (Y5+): platform licensing 50-60%, per-use services 15-25%, custom modeling 10-15%, grants/partnerships 5-10%, outcome-based 5-10%.

15. Funding Path

Stage	Amount	Gate
Bootstrap (Mo 1-6)	\$0	Prototype + preprint (OLIVES)
Pre-seed / Grants (Mo 6-12)	\$50K-\$300K	NIH SBIR, NEI, ARPA-H (independent-eligible)
Seed (Mo 12-18)	\$500K-\$2M	Published + scale-data path + partnership interest
Series A (Y2-3)	\$5M-\$15M	Scale validation + pilot
Series B (Y3-4)	\$20M-\$50M	Multi-disease + revenue
Series C / Exit (Y5-7)	\$50M+ / acq.	Category leader + revenue

Non-dilutive first. SBIR/STTR programs fund independent companies — no university required. NIH SBIR Phase I (~\$300K) easily covers UK Biobank fees and compute. Pursue NEI interest areas, ARPA-H, American Diabetes Association, Breakthrough T1D before venture capital. Preserve ownership; demonstrate validation.

16. IP Strategy (Gap Covered)

If we publish openly, the proprietary layer must be explicit. Our defensible IP is NOT the published methods — it is:

Proprietary data assets. Curated longitudinal datasets and partnership-exclusive access compound over years

Trained model checkpoints. Models trained on proprietary controlled data are not reproducible from the paper alone

Validation + regulatory credibility. Years of clinician-adjudicated validation are hard to replicate

Integrations + switching costs. Workflow and pharma-pipeline integrations create lock-in

Trade secrets. Specific engineering, calibration, and fusion techniques kept unpublished

Open vs closed: Publish benchmarks, baselines, and public-data methods (credibility). Keep closed: proprietary-data checkpoints, fusion engineering, and anything that exposes controlled-data-derived capability. **If UW or any institution is involved, settle IP ownership upfront via their tech-transfer office BEFORE using their resources or data.**

17. Runway Math (Gap Covered)

The founder has no dedicated runway and builds alongside other commitments. The plan is engineered for near-zero burn through Month 6:

Item	Cost	Notes
Compute (Mo 1-6)	\$0-50/mo	Google Colab free tier; Colab Pro if needed
Data (Mo 1-6)	\$0	OLIVES, EyePACS, Messidor, FGADR all free
Tools	\$0-50/mo	GitHub free, W&B free tier, AI assistants
UK Biobank (when ready)	~£9K (~\$11K)	Fund via SBIR, not personal runway
Total burn Mo 1-6	< \$500	Time is the real cost, not cash

Implication: Months 1-6 cost almost nothing but time, so financial pressure should not force premature decisions. The first cash need (UK Biobank, compute scale) is timed to coincide with grant funding. Keep income stable; do not quit anything until evidence justifies it.

Part VI — Risk & Decision Gates

18. Risk Register

Scientific

Risk	Sev	Mitigation
OLIVES too small for strong claim	HIGH	Scope claim to prototype; controlled data for scale
Models learn artifacts not disease	HIGH	External validation; subgroup analysis; clinician review
Disease dynamics not learnable	HIGH	Falsify and pivot — accept this is possible
OLIVES temporal depth too thin	MED	Feasibility audit FIRST; adjust model class accordingly

Data Access

Risk	Sev	Mitigation
All controlled paths stall	MED	Multiple parallel routes; OLIVES sustains Y1 regardless
UK Biobank cost	LOW-MED	Grant funding; not personal runway
Partnership cycles slow	MED	Start Month 1-3; don't gate Y1 on them

Business

Risk	Sev	Mitigation
Pharma sales cycles too slow	HIGH	Non-dilutive bridge; grants early
Well-resourced labs replicate	MED	Data + validation moats, not just publications
Regulatory shifts	MED	Position as research tool initially

Personal

Risk	Sev	Mitigation
Competing commitments dilute focus	HIGH	Sustainable cadence; AI leverage; protect deep-work blocks
Burnout	HIGH	Realistic pace; celebrate rung-by-rung progress
Isolation reduces quality	MED	Community engagement; advisor cultivation
IP entanglement if affiliated	MED	Settle ownership upfront via tech-transfer

19. Decision Gates (Pre-Committed)

We do not move these gates to avoid hard decisions. Each has three outcomes: CONTINUE / ITERATE / PIVOT.

Gate 1 — Month 3

Question: Infra + baselines work + OLIVES audited?

Continue if: Baselines run; clean repo; OLIVES feasibility known; UK Biobank app in

Gate 2 — Month 6

Question: Architecture functions on real temporal data?

Continue if: Encoder beats baselines; OLIVES dynamics prototype works; ≥ 1 scale-data path moving

Gate 3 — Month 12

Question: Earned the right to pursue seriously?

Continue if: Preprint published; scale-data path approved/late-stage; funding interest

Gate 4 — Month 18

Question: Real data confirms thesis at scale?

Continue if: Validated on controlled longitudinal data; treatment-aware working; first revenue/grant

Gate 5 — Month 24

Question: Real business or research project?

Continue if: First paid pilot; multimodal working; team forming; Series A discussions

Gate 6 — Month 36

Question: Platform thesis working?

Continue if: Multi-disease; pharma partnerships; clear path to scale

The willingness to PIVOT or FOLD at a failed gate is what separates discipline from delusion. Pivot options at each stage: narrower progression-prediction product, academic/research tool, data-infrastructure collaboration, or graceful wind-down.

20. The Discipline & The Mission

This document is alive; it will be revised as evidence accumulates. Every tactical document — sprint plans, OKRs, technical specs, data playbooks — references back here.

Hold Two Thoughts At Once

***First:** We are building something genuinely valuable to medicine — infrastructure for understanding how disease evolves through time. The vision is real and worth pursuing.*

***Second:** We have not yet earned the right to claim we will succeed. The science may not hold. Every claim must be evidence-backed; every uncertainty honest.*

The Critical Path

1. Build the OLIVES + public prototype — starts today, solo, near-zero cost.
2. Secure controlled-data access via multiple independent routes (no single dependency).
3. Validate dynamics at scale (Year 2).
4. Everything else — JEPA, HAWVA, simulation, synthetic cohorts, pharma — follows.

The Mission

Build disease dynamics. Earn the right through evidence. Depend on no single institution. Save lives. And if the science holds and the data flows — build the unicorn.

The single most important near-term action: Download OLIVES and run the feasibility audit. It is the first brick of the disease-dynamics engine, it requires nothing but time, and it tells us whether the architecture has real temporal structure to learn from. Everything starts there.